Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________

Even, Odd, or Neither

Example (Determine even, odd, or neither):
$f(x) = x^4 - x^2$: $f(-x) = (-x)^4 - (-x)^2 = x^4 - x^2 = f(x)$, so it is **even**.

Directions: Compute $f(-x)$ and simplify, then classify as even, odd, or neither.

1. $f(x) = x^2 + 1$

2. $f(x) = x^3 - x$

3. $f(x) = x^3 + 2$

4. $f(x) = |x|$

5. $f(x) = 2x$

6. $f(x) = x^4 - 3x^2$

7. $f(x) = x^5$

8. $f(x) = x^2 + x$

9. $f(x) = \dfrac{1}{x}$

10. **(challenge)** $f(x) = x^3 + x^2$

Challenge (same sheet):

11. **(challenge)** Explain why any function made ONLY of even-power terms (like $x^4 - 3x^2 + 5$) must always be even, using the substitution rule.

Answer Key — fully worked

Procedure: Replace every x in the function with $-x$.; Simplify the result completely.; If $f(-x) = f(x)$, the function is even.; If $f(-x) = -f(x)$, the function is odd.; If neither matches, the function is neither..

1. $f(-x) = x^2 + 1 = f(x)$: even

2. $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$: odd

3. $f(-x) = -x^3 + 2$, which is neither $f(x)$ nor $-f(x)$: neither

4. $f(-x) = |-x| = |x| = f(x)$: even

5. $f(-x) = -2x = -f(x)$: odd

6. $f(-x) = x^4 - 3x^2 = f(x)$: even (all even-power terms)

7. $f(-x) = -x^5 = -f(x)$: odd

8. $f(-x) = x^2 - x$, which is neither $f(x)$ nor $-f(x)$: neither

9. $f(-x) = \dfrac{1}{-x} = -\dfrac{1}{x} = -f(x)$: odd

10. $f(-x) = -x^3 + x^2$: neither — the even x^2 term stays positive while the odd x^3 term flips sign, so it can't equal $f(x)$ or $-f(x)$

11. When you substitute $-x$ into any even-power term, the negative disappears: $(-x)^2 = x^2$, $(-x)^4 = x^4$, and in general $(-x)^{2n} = x^{2n}$ because a negative base raised to an even power is positive. A constant term (like $+5$) is also unchanged. So if EVERY term has an even power (or is a constant), then $f(-x)$ equals the original $f(x)$ term-for-term. For example, $f(x) = x^4 - 3x^2 + 5$ gives $f(-x) = (-x)^4 - 3(-x)^2 + 5 = x^4 - 3x^2 + 5 = f(x)$. Since $f(-x) = f(x)$, the function must be even.